
```
clc;
clear;
close all;
```

Parameters

```
M = 16;           % 16-QAM
N = 1000;         % Number of symbols
SNR = 10;         % SNR in dB
```

Generate Random Data

```
data = randi([0 M-1], N, 1);
```

16-QAM Modulation

```
tx = qammod(data, M, 'UnitAveragePower', true);
```

AWGN Channel

```
rx = awgn(tx, SNR, 'measured');
```

16-QAM Demodulation

```
data_rx = qamdemod(rx, M, 'UnitAveragePower', true);
```

Convert Symbols to Bits

```
tx_bits = de2bi(data, 4, 'left-msb');
rx_bits = de2bi(data_rx, 4, 'left-msb');
```

```
tx_bits = tx_bits(:);
rx_bits = rx_bits(:);
```

BER Calculation

```
[num_errors, ber] = biterr(tx_bits, rx_bits);
```

Display BER

```
fprintf('Number of Bit Errors = %d\n', num_errors);
fprintf('Bit Error Rate (BER) = %.6f\n', ber);
```

```
Number of Bit Errors = 249
Bit Error Rate (BER) = 0.062250
```

Eye Diagram Signal

```
eyeSig = repelem(real(rx), 8);
```

Plot Results

```
figure('Name','16-QAM Modulation','NumberTitle','off');

% Transmitted Constellation
subplot(2,2,1);
scatter(real(tx), imag(tx), 20, 'filled');
title('16-QAM Transmitted Constellation');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis equal;

% Received Constellation
subplot(2,2,2);
scatter(real(rx), imag(rx), 20, 'filled');
title('Received Constellation (AWGN)');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis equal;

% Eye Diagram
subplot(2,2,[3 4]);
eyediagram(eyeSig, 16);
title('Eye Diagram');
```

